INCH-POUND
MIL-DTL-24643/5F
22 August 2002
SUPERSEDING
MIL-C-24643/5E
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE LSMDU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First -	Copper conductor, uncoated (see table I for size). Coated copper may be used at manufacturer's option where required to provide free stripping insulation.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Ethylene propylene rubber or cross-linked polyethylene insulation or at the manufacturer's option a composite with a primary insulation consisting of ethylene propylene rubber or cross-linked polyethylene. The composite layer may be substituted for up to 10 percent of the wall thickness (see table I for wall thickness). Standard identification code shall be applied by method 1 or method 3. If identification method 1 is used, insulation color shall be black.
Fourth -	Nineteen conductors cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence to be consecutive, starting with No. 1 from center outward. Fillers shall be used as necessary to make a firm, well rounded assembly.
Fifth -	Binder tape applied helically with overlap.
Sixth -	Cross-linked polyolefin jacket. (see table I for wall thickness)

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TABLE I. Details.

Military part no. M24643/5	Type and size	Conductors AWG size	Insulation thickness minimum average (inch)	Cable jacket thickness minimum average (inch)	Overall diameter maximum (inch)	Conductor resistance per 1000 feet maximum (ohms)	Insulation resistance for 1000 feet minimum (megohms)	Cold bending mandrel diameter max (inch)
-01UN	LSMDU-6	12 (class B)	0.028	0.060	1.000	1.730	100	13
-02UN	LSMDU-14	9 (class B)	0.040	0.060	1.395	0.868	100	19
-03UN	LSMDU-23	7 (class B)	0.052	0.075	1.765	0.598	100	23
-04UN	LSMDU-40	4 (class C)	0.052	0.075	2.040	0.273	90	27
-05UN	LSMDU-60	2 (class D)	0.052	0.075	2.330	0.172	75	30

EXAMINATION AND TESTS:

Basic electrical: Requirements:

Conductor resistance - ohms/1000 feet at 25°C, maximum .....(see table I)
Voltage withstand - volts, root mean square, minimum
Conductor to conductor 2500
Insulation resistance - megohms -1000 feet, minimum .....(see table I)
Conductor continuity ..... No failure

Group A:

Visual and dimensional..... No failure
Watertightness – see MIL-DTL-24643 for limits of water leakage ..... No failure

Group B:

Cold bending, cable – (see table I for mandrel diameter)..... No damage
Drip – 95 ± 1°C.....Zero
Cross-linked proof test (percent, maximum)
Insulation 50
Jacket (When tested at 200°C).....50
Tear - pounds per inch thickness, minimum (ASTM D 470) 35

Physicals (unaged)

Insulation
Ethylene propylene rubber
Tensile strength - lb/in^2, minimum 700
Elongation - percent, minimum. 250
Cross-linked polyethylene
Tensile strength - lb/in^2,minimum 1800
Elongation - percent, minimum 250
Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160

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Group C:
Physicals (aged)
- Insulation
  - Ethylene propylene
    - Air oven
      - Tensile strength - percent of unaged, minimum 75
      - Elongation - percent of unaged minimum 75
  - Cross-linked polyethylene
    - Air oven
      - Tensile strength - percent of unaged, minimum 80
      - Elongation - percent of unaged minimum 80
  - Jacket (cable)
    - Air oven
      - Tensile strength - percent of unaged, minimum 60
      - Elongation - percent of unaged minimum 60
  - Hot oil immersion
    - Tensile strength - percent of unaged, minimum 50
    - Elongation - percent of unaged minimum 50
- Shrinkage ..... No failure
- Permanence of printing (jacket) - cycles, minimum 125
- Permanence of printing (conductor) - cycles, minimum 25
- Heat distortion - percent of unaged, maximum 30
- Cable sealant removability ..... No failure

Group D:
- Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Requirements:
- Aging and compatibility (cable) (125 ± 5°C) ..... No failure
- Abrasion resistance (jacket) - scrapes, minimum 75
- Acid gas equivalent - percent, maximum
  - Jacket 2
  - Fillers 2
  - Insulation 18
- Halogen content - percent, maximum
  - Jacket 0.2
  - Fillers 0.2
- Immersion (jacket)
  - Tensile strength - percent of unaged, minimum 50
  - Elongation - percent of unaged, minimum 50

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Smoke index, maximum
- Jacket 25
- Fillers 45
- Insulation 45

Toxicity index, maximum
- Jacket 5
- Fillers 5
- Insulation 1.5

Durometer (jacket) - (type A) hardness, minimum.....80
Weathering (jacket)..... No failure

UNIT ORDERING LENGTH: 1000 feet nominal length for sizes 6, 14, and 23; 500 feet nominal for sizes 40 and 60.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-004)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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